

DATE AND TIME JAVA 8 :

THERE ARE 8 CLASSES OF IT.

- LOCAL DATE
- LOCAL TIME
- LOCAL DATE TIME
- ZONED DATE TIME
- INSTANT
- DURATION
- PERIOD
- DATE TIME FORMATTER.

CODE FOR ALL:

```
package ex;

import java.time.*;
import java.time.format.DateTimeFormatter;
import java.util.Set;

/*
1.LocalDate
2.LocalTime
3.LocalDateTime
4.ZonedDateTime
5.Instant
6.Period
7.Duration
8.DateTimeFormatter
*/
public class Demo {
    public static void main(String[] args){

        // LOCAL DATE

        // way to create local date with custom
        LocalDate customDate = LocalDate.of(2025, 7, 18);
        System.out.println(customDate);

        // way to create local date with system date
```

```

LocalDate nowDate = LocalDate.now();
System.out.println(nowDate);

// these are the methods can be applied on above created date
int dayOfMonth = nowDate.getDayOfMonth();
int year = nowDate.getYear();
Month month = nowDate.getMonth();
int value = nowDate.getMonth().getValue();
System.out.println(dayOfMonth);
System.out.println(year);
System.out.println(month);
System.out.println(value);

LocalDate yesterday = nowDate.minusDays(1);
System.out.println(yesterday);

if (nowDate.isAfter(yesterday)) {
    System.out.println("ha shi hai");
}

// LOCAL TIME

// way to create local time with custom time
LocalTime customTime = LocalTime.of(14, 30, 45);
System.out.println(customTime);

// way to create local time with system time
LocalTime nowTime = LocalTime.now();

// if time in string we can parse it
String strTime="13:20:13";
LocalTime parse = LocalTime.parse(strTime);
System.out.println(parse);

// like this below method we can use lots of method similar to local date
LocalTime minusHours = nowTime.minusHours(1);
System.out.println(minusHours);

// LOCAL DATE TIME
// way to create local Datetime with custom Datetime
LocalDateTime customDateTime = LocalDateTime.of(2024, 12, 30, 13, 0, 0);
System.out.println(customDateTime);

// we can parse it and perform above methods also.

// ZONED DATE TIME

// way to create custom zone date time
ZonedDateTime customeZonedDateTime = ZonedDateTime.of(2000, 12, 1, 12,

```

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20, 25, 10, ZoneId.of("Australia/Hobart"));
System.out.println(customZonedDateTime);

// way to create current zoned date time with system
ZonedDateTime zonedDateTime = ZonedDateTime.now();
Set<String> availableZoneIds = ZoneId.getAvailableZoneIds();
System.out.println(availableZoneIds.size());
availableZoneIds.forEach(t-> System.out.println(t));

// lots of operation we can perform on it like above .

// USED FOR MILLISECOND
Instant now1 = Instant.now();
System.out.println(now1);

Instant now2 = Instant.now();

// USED FOR COMPARING with minutes & seconds
Duration duration = Duration.between(now1, now2);
System.out.println(duration);

// PERIOD FOR COMPARING WITH YEARS, MONTH, DAY
LocalDate now = LocalDate.now();
LocalDate then = LocalDate.of(1982, 3, 4);
Period between = Period.between(nowDate, then);
System.out.println(between);

// DATE TIME FORMATTER
LocalDate localDate = LocalDate.now();
System.out.println(localDate);
DateTimeFormatter dateTimeFormatter =
DateTimeFormatter.ofPattern("dd/MM/yyyy");
String format = localDate.format(dateTimeFormatter);
System.out.println(format);

String date = "25/04/1998";
DateTimeFormatter dtFormatter =
DateTimeFormatter.ofPattern("dd/MM/yyyy");
LocalDate parsedStringDate = LocalDate.parse(date, dtFormatter);
String formatted = parsedStringDate.format(dtFormatter);
System.out.println(formatted);

String dateTimeString = "2023-04-24 10:30:45 IST";
DateTimeFormatter dateTimeFormatter1 =
DateTimeFormatter.ofPattern("yyyy-MM-dd HH:mm:ss z");
ZonedDateTime parse1 = ZonedDateTime.parse(dateTimeString,
dateTimeFormatter1);
System.out.println(parse1);
}

```

}